Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Applications of Systems (Word Problems)

Name: ______________
Date: __________

Quiz (Habanero)

Q1. A city's temperature t hours after sunrise is given by $t = 2x + 5$ and the temperature t hours before sunset is $t = 3x - 2$. What is the temperature at sunrise and sunset if $x = 2$?
(A) 9°C and 4°C
(B) 10°C and 5°C
(C) 11°C and 6°C
(D) 12°C and 7°C
(E) 13°C and 8°C

Q2. The amount of rainfall r in a region is related to the amount of forest cover f by the equation $r = 0.5f + 10$. If the region has 20
(A) 16 mm
(B) 17 mm
(C) 18 mm
(D) 19 mm
(E) 20 mm

Q3. A researcher is studying the effect of pollution on the ecosystem. Let x be the amount of waste and y be the number of species. The equations are $2x + 3y = 12$ and $x - 2y = -3$. How many species will there be when there is 3 units of waste?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q4. A weather balloon is rising at a rate of $2 \cdot t$ m/s, and its temperature T is given by $T = 20 - 0.5t$. If the initial temperature is $15C$, how long will it take for the temperature to drop by $5C$?
(A) 5 s
(B) 6 s
(C) 7 s
(D) 8 s
(E) 9 s

Q5. The cost c of a project to reduce pollution is related to the amount of pollution reduced p by the equation $c = 2p + 1000$. If the project reduces pollution by 15
(A) 1200
(A) 1300
(A) 1400
(A) 1500
(A) 1600

Q6. A region has a temperature range given by $t = x^2 - 4x + 10$ and $t = -x^2 + 6x - 5$. What is the maximum temperature if x represents the time of day?
(A) 11°C
(B) 12°C
(C) 13°C
(D) 14°C
(E) 15°C

Q7. A water tank can hold 1000 gallons of water. The amount of water w in gallons and the time t in hours are related by $w = 50t + 200$ and $w = -20t + 800$. How long will it take for the tank to fill?
(A) 2 h
(B) 4 h
(C) 6 h
(D) 8 h
(E) 10 h

Q8. A city has a population growth rate given by $p = 2x + 5000$ and a population decrease rate due to migration given by $p = -x + 2000$. What is the equilibrium population if x represents the number of years?
(A) 3000
(B) 4000
(C) 5000
(D) 6000
(E) 7000

Open Ended Questions (FRQ)

Q9. Suppose a weather forecasting model gives the temperature t as a function of time x as $t = 3x - 2$ and $t = x^2 - 4$. If x represents the time of day, find the times when the temperature is the same for both models and determine the temperature at those times. Show your work and justify your answer.

Q10. A park ranger is monitoring the ecosystem of a forest. The population of rabbits r and the population of foxes f are related by the equations $2r + 3f = 12$ and $r - 2f = -1$. If the current population of rabbits is 2, find the population of foxes and determine if the ecosystem is stable. Justify your answer and provide any relevant calculations.

Chef's Tips

- Read each question carefully
- Use a pencil to fill in the answer sheet
- Check your work before submitting